BADR UNIVERSITY IN CAIRO (BUC) — FACULTY OF ENGINEERING

Fluid Power Engineering (Dr. M. Galal Rabie) — Chapter 1 Classic Solved Problems

Course: MEC 431 / MEC 451 | **Examiner:** Dr. Ziad Ahmed Ibrahim | **Teaching Assistant:** Eng. Moh Zedan

Detailed Pedagogical Solutions & Physical Explanations (Problems 6, 7, 8, and 9)

Problem 6: Standard Extension Mode of a Hydraulic Cylinder

Problem Statement:
The figure shows the standard extension mode of a hydraulic cylinder. Neglecting line and valve losses, the loading force is $F = 20,000$ N.
Given: Pump flow rate $Q_P = 40$ L/min, Piston diameter $D = 100$ mm $= 0.1$ m, Piston rod diameter $d = 70$ mm $= 0.07$ m.
Calculate: Return flow rate Q_T, piston velocity v, cylinder mechanical output power N_m, pump hydraulic output power N_h, and system pressure p. Comment on the results.

Step-by-Step Mathematical Solution:

1. Cylinder Cross-Sectional Areas:
Full piston area:

$$A_p = \frac{\pi}{4}D^2 = \frac{\pi}{4}(0.1)^2 = \mathbf{0.007854\ m^2} = 78.54\ \text{cm}^2$$

Piston rod area:

$$A_{rod} = \frac{\pi}{4}d^2 = \frac{\pi}{4}(0.07)^2 = 0.003848\ \text{m}^2$$

Annular (rod-end) area:

$$A_r = A_p - A_{rod} = \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}[(0.1)^2 - (0.07)^2] = \mathbf{0.004006\ m^2} = 40.06\ \text{cm}^2$$

2. Piston Extension Speed (v):
Convert pump flow to SI: $Q_P = \frac{40}{60,000} = 6.667 \times 10^{-4}\ \text{m}^3/\text{s}$.

$$v = \frac{Q_P}{A_p} = \frac{6.667 \times 10^{-4}}{0.007854} = \mathbf{0.08488\ m/s} = \mathbf{8.49\ cm/s}$$

3. Returned Flow Rate to Reservoir (Q_T):

$$Q_T = v \cdot A_r = 0.08488 \times 0.004006 = 3.400 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{20.40\ L/min}$$

4. System Pressure Required to Overcome Load (p):

$$p = \frac{F}{A_p} = \frac{20,000\ \text{N}}{0.007854\ \text{m}^2} = 2,546,479\ \text{Pa} = \mathbf{2.55\ MPa = 25.46\ bar}$$

5. Power Calculations:
Mechanical Output Power:

$$N_m = F \cdot v = 20,000 \times 0.08488 = \mathbf{1698\ W = 1.70\ kW}$$

Pump Hydraulic Power:

$$N_h = Q_P \cdot p = (6.667 \times 10^{-4}) \times (2,546,479) = \mathbf{1698\ W = 1.70\ kW}$$

Summary & Engineering Comment:

Speed: **v = 8.49 cm/s** | Return Flow: **Q_T = 20.40 L/min** | Pressure: **p = 25.46 bar**

Powers: **N_m = N_h = 1698 W**

Comment: Mechanical output power exactly equals hydraulic input power because transmission friction and valve throttling losses were neglected ($\eta = 100\%$).

Problem 7: Differential (Regenerative) Connection Mode

Problem Statement:
The cylinder from Problem 6 is now connected in **differential connection** (both cap end and rod end connected to the pump line). Line pressure is $P = 200\ \text{bar} = 20\ \text{MPa}$. Pump delivery $Q_P = 40\ \text{L/min}$. Neglecting losses, calculate: loading force F, inlet flow Q_{in}, returned flow Q_{out}, speed v, N_m, and N_h. Compare with Problem 6.

Step-by-Step Mathematical Solution:

1. Physical Principle & Velocity Derivation:
Fluid displaced from the rod end cannot return to tank; it recirculates directly into the cap end:

$$Q_{in} = Q_P + Q_{out} \implies v \cdot A_p = Q_P + v \cdot A_r \implies v(A_p - A_r) = Q_P$$

Since $A_p - A_r = A_{rod} = \frac{\pi}{4}d^2$:

$$v = \frac{Q_P}{A_p - A_r} = \frac{Q_P}{A_{rod}} = \frac{6.667 \times 10^{-4}}{\frac{\pi}{4}(0.07)^2} = \frac{6.667 \times 10^{-4}}{0.003848} = \mathbf{0.17326\ m/s = 17.33\ cm/s}$$

2. Total Flow Entering Cap End (Q_{in}) and Regenerated Flow (Q_{out}):

$$Q_{in} = v \cdot A_p = 0.17326 \times 0.007854 = 1.3608 \times 10^{-3}\ \text{m}^3/\text{s} = \mathbf{81.65\ L/min}$$

$$Q_{out} = v \cdot A_r = 0.17326 \times 0.004006 = 6.941 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{41.65\ L/min}$$

(Check: $Q_{in} - Q_{out} = 81.65 - 41.65 = 40.00\ \text{L/min} = Q_P$ — continuity is perfectly satisfied!).

3. Maximum Loading Force at $P = 200\ \text{bar} = 20\ \text{MPa}$:
Net thrust force is the difference between cap end thrust and rod end opposition:

$$F = P \cdot A_p - P \cdot A_r = P(A_p - A_r) = P \cdot A_{rod}$$

$$F = (20 \times 10^6\ \text{Pa}) \times (0.003848\ \text{m}^2) = 76,969\ \text{N} = \mathbf{76.97\ kN} \approx \mathbf{77\ kN}$$

4. Power Calculations:

$$N_h = Q_P \cdot P = (6.667 \times 10^{-4}) \times (20 \times 10^6) = \mathbf{13,333\ W = 13.33\ kW}$$

$$N_m = F \cdot v = 76,969 \times 0.17326 = \mathbf{13,336\ W = 13.34\ kW}$$

Direct Comparison Table:

Performance Parameter	Standard Extension (Problem 6)	Differential Mode (Problem 7)	Physical Effect
Piston Speed (v)	8.49 cm/s	**17.33 cm/s**	**2.04× Faster** (A_p/A_{rod})
Max Thrust Force (F)	$P \cdot A_p = 157.1$ kN	$P \cdot A_{rod} = \mathbf{76.97\ kN}$	**2.04× Lower** thrust
Cap End Inflow (Q_{in})	40.0 L/min	**81.65 L/min**	More than 2× pump delivery
Fluid Discharged to Tank	20.40 L/min	**0.0 L/min**	Bypasses tank completely
Power Efficiency	100%	100%	Acts as an ideal hydraulic gear ratio

Problem 8: Load-Lowering System with Meter-Out Speed Control

Problem Statement:
In a load-lifting system, the lowering speed is controlled by a **throttle-check valve** connected in the rod-end line (meter-out).
Given Data: Overhung weight $W = 30\ \text{kN} = 30,000\ \text{N}$, Pump exit pressure $P = 30\ \text{bar} = 3 \times 10^6\ \text{Pa}$,
Piston speed during lowering $v = 0.07\ \text{m/s}$, Cap area $A_p = 78.5\ \text{cm}^2 = 0.00785\ \text{m}^2$, Rod area $A_r = 40\ \text{cm}^2 = 0.004\ \text{m}^2$,
Oil density $\rho = 870\ \text{kg/m}^3$, Discharge coefficient $C_d = 0.611$, Safety relief valve setting $= 350\ \text{bar}$.
Calculate: Pressure in cylinder rod side (P_c), inlet flow (Q_{in}), outlet flow (Q_{out}), pump flow (Q_P), pump power (N_h), and throttle area (A_t).

Step-by-Step Mathematical Solution:

1. Flow Rates in Lowering Mode:
Piston extends downward at $v = 0.07\ \text{m/s}$:
Flow entering cap end:

$$Q_{in} = Q_P = v \cdot A_p = 0.07 \times 0.00785 = 5.495 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{32.97\ L/min}$$

Flow escaping rod end through throttle valve:

$$Q_{out} = v \cdot A_r = 0.07 \times 0.00400 = 2.800 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{16.80\ L/min}$$

2. Static Force Balance to Calculate Rod-End Pressure (P_c):
Downward forces (Gravity load W + Pump pressure thrust $P \cdot A_p$) are resisted by the throttled backpressure $P_c \cdot A_r$:

$$P_c \cdot A_r = W + P \cdot A_p$$

$$P_c = \frac{W + P \cdot A_p}{A_r} = \frac{30,000 + (3 \times 10^6 \times 0.00785)}{0.004}$$

$$P_c = \frac{30,000 + 23,550}{0.004} = \frac{53,550}{0.004} = 13,387,500\ \text{Pa} = \mathbf{13.39\ MPa = 133.9\ bar}$$

3. Sizing the Throttle Valve Area (A_t):
Fluid discharges from P_c through the throttle into the reservoir ($P_{tank} = 0$):

$$Q_{out} = C_d A_t \sqrt{\frac{2P_c}{\rho}} \implies A_t = \frac{Q_{out}}{C_d \sqrt{\frac{2P_c}{\rho}}}$$

$$\sqrt{\frac{2 \times 13.3875 \times 10^6}{870}} = \sqrt{30,775.86} = 175.43\ \text{m/s}$$

$$A_t = \frac{2.800 \times 10^{-4}}{0.611 \times 175.43} = \frac{2.800 \times 10^{-4}}{107.188} = 2.612 \times 10^{-6}\ \text{m}^2 = \mathbf{2.61\ mm^2}$$

4. Pump Hydraulic Output Power (N_h):

$$N_h = P \cdot Q_P = (3 \times 10^6\ \text{Pa}) \times (5.495 \times 10^{-4}\ \text{m}^3/\text{s}) = \mathbf{1648.5\ W \approx 1.65\ kW}$$

Summary of Results for Problem 8:

Rod-Side Backpressure: **P_c = 13.39 MPa (133.9 bar)** | Throttle Area: **A_t = 2.61 mm²**

Pump Flow: **Q_p = 32.97 L/min** | Pump Power: **N_h = 1.65 kW**

Problem 9: Lifting Mode & Maximum Capacity of Load-Lifting System

Problem Statement:
Redraw the circuit of Problem 8 in **lifting mode** (cylinder retracts upward to hoist load W).
For the same pump delivery ($Q_P = 32.97\ \text{L/min}$) and safety relief valve setting ($P_r = 350\ \text{bar} = 35\ \text{MPa}$), calculate:
(a) Lifting speed v.
(b) Maximum load W_{max} that the system can lift.
(c) Operating parameters and pump power N_h.

Step-by-Step Mathematical Solution:

1. Flow Routing in Lifting Mode:
The 4/3 DCV shifts to cross-over position:

- Pump flow Q_P enters the rod end (A_r) through the free-reverse check valve of the throttle-check assembly (bypassing the throttle orifice with zero resistance!).
- Oil expelled from the cap end (A_p) discharges through the DCV back to tank at 0 bar.

2. Lifting Speed (v):

$$v = \frac{Q_P}{A_r} = \frac{5.495 \times 10^{-4}\ \text{m}^3/\text{s}}{0.00400\ \text{m}^2} = \mathbf{0.1374\ m/s = 13.74\ cm/s}$$

(Notice: Retraction speed is almost twice as fast as lowering speed: $13.74\ \text{cm/s}$ *vs* $7.0\ \text{cm/s}$*).*

3. Maximum Lifting Capacity (W_{max}):
Maximum lifting force occurs when pump pressure reaches the relief valve setting $P_r = 350\ \text{bar} = 35 \times 10^6\ \text{Pa}$:

$$W_{max} = P_r \cdot A_r = (35 \times 10^6\ \text{Pa}) \times (0.00400\ \text{m}^2) = 140,000\ \text{N} = \mathbf{140\ kN}$$

In equivalent metric tons:

$$m_{max} = \frac{140,000}{9.81} = \mathbf{14,271\ kg \approx 14.27\ metric\ tons}$$

4. Pump Hydraulic Power at Maximum Lift (N_h):

$$N_h = P_r \cdot Q_P = (35 \times 10^6\ \text{Pa}) \times (5.495 \times 10^{-4}\ \text{m}^3/\text{s}) = 19,232.5\ \text{W} = \mathbf{19.23\ kW}$$

Summary of Results for Problem 9:

Lifting Velocity: **v = 13.74 cm/s** | Maximum Load Capacity: **W_max = 140 kN (14.27 tons)**

Cap End Return Flow: **Q_out = 64.67 L/min** | Maximum Hydraulic Power: **N_h = 19.23 kW**